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Intelligent Data Processing and Applications

COE 543/743

Chapter 9 - Semantic Disambiguation

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Computer Software Engineering

Ch. 9 – Semantic Disambiguation

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1. Introduction
2. Brief History
3. Applications
4. Traditional WSD
5. WSD for Semi-structured Data
6. Challenges and Ongoing Directions

Ch. 9 – Semantic Disambiguation

1. Introduction

- Word Sense Disambiguation(WSD) is the task of assigning the correct meaning to a word given its context
 - Since a word can have multiple meanings: **polysemy**
- Example:
 - The word **date** can be assigned different meanings:
 1. Larry took Daisy out on a **date**
 2. Larry's favorite fruit to eat is a **date**
 - In the first sentence, the meaning of the word date is a *social engagement*
 - In the second sentence, the word date is a *fruit*

Ch. 9 – Semantic Disambiguation

1. Introduction

- Humans are quite good when it comes to WSD, ranking above 90% accuracy
 - Machine WSD accuracy remains lower than human WSD accuracy
- Replicating this human ability using algorithms can be quite a demanding and difficult task
- Solving this problem benefits many computer-related applications
 - Such as improving search engine relevance, speech recognition, and language translations



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2. Brief History

- WSD is one of the oldest problems in computation linguistics that dates back to the 40s
- Introduced by **Warren Weaver**, a pioneer of the machine translation
- WSD was believed, by early researchers, to be unsolvable by electronic computers due to the necessity of having to model all knowledge
- In the 1970, WSD systems were still very primitive as they were:
 - Hand coded
 - Rule based (if-then statements)



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2. Brief History

- In the 1980s, the emergence of large-scale lexical resources enabled the switch from hand-coded knowledge to automatic knowledge extraction from these resources
- In the 1990s, knowledge-based solutions helped improve of WSD
 - Corpus-based techniques were picking up
- In the 2000s, knowledge-based and supervised learning solutions reached human-like accuracy levels, and other methods emerged such as domain adaptation, semi-supervised, and unsupervised word embedding solutions
- Hybrid systems are currently trending, combining knowledge-based and corpus-based techniques

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3. Applications

- WSD is not a standalone application
 - It is an **intermediate task** which is not an end in itself
 - Necessary to accomplish most NLP tasks
 - Machine translation
 - Information retrieval
 - Content and thematic analysis
 - Grammatical analysis
 - Named entity resolution
 - Text classification
 - ...

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3. Applications

- **Machine Translation:** Words that have different translations for different senses
 - *Example:* Translate the French word “grille”
 - In English, it can be: railings, gate, bar, grid, scale, schedule, etc.
 - *Example:* Translate the English word “change”
 - In French, it can be: *changement* ('transformation') or *monnaie* ('pocket money')
 - *Example:* Translate the Italian word “penna”
 - In English, it can be: *feather, pen or author*

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3. Applications

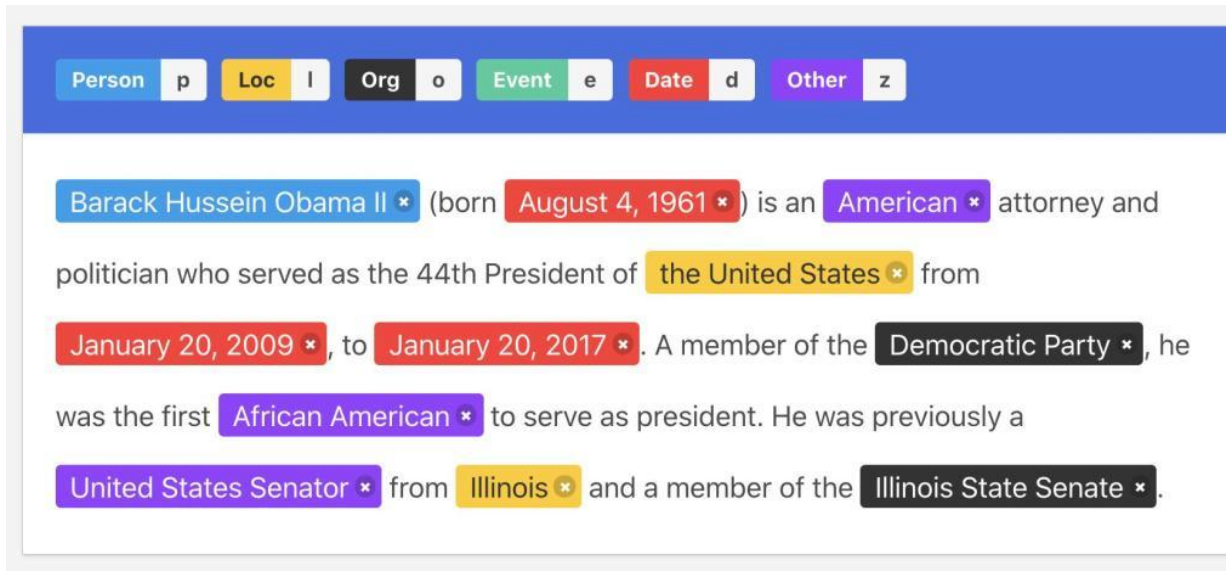
- **Information Retrieval:** The need to eliminate occurrences in documents where the words are used in an inappropriate sense
 - *Example:* Find all Web Pages about “depression”
 - The illness, weather systems, or economics!?
 - *Example:* Find all Web Pages about “cricket”
 - The sport or the insect?



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3. Applications

- **Named entity resolution:** the classification of target textual items into predefined categories
 - Using a predefined set of parts of speech classes

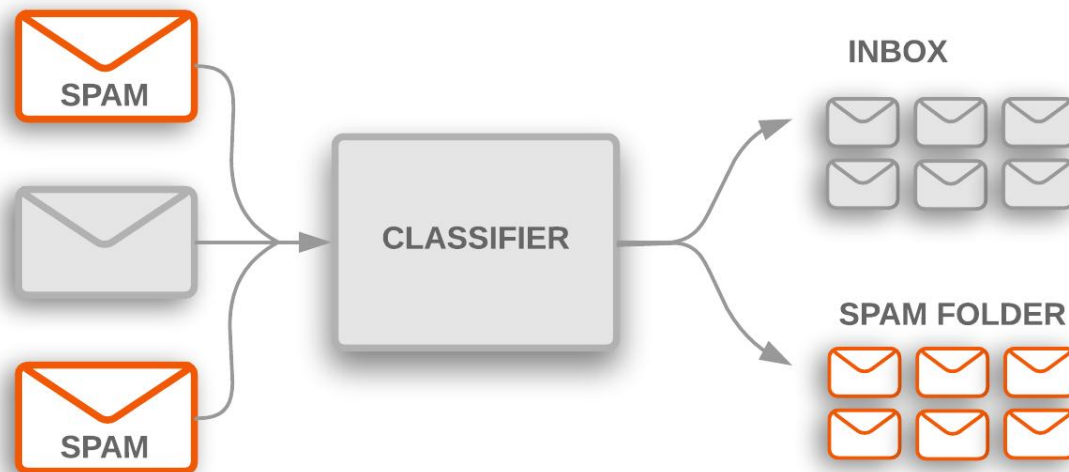


The screenshot displays a Named Entity Recognition (NER) interface. At the top, there is a legend bar with colored boxes and labels: Person (p), Loc (l), Org (o), Event (e), Date (d), and Other (z). Below the legend, a text snippet is shown with various entities highlighted in colored boxes and labeled with the corresponding letter from the legend. The text is: "Barack Hussein Obama II (born August 4, 1961) is an American attorney and politician who served as the 44th President of the United States from January 20, 2009, to January 20, 2017. A member of the Democratic Party, he was the first African American to serve as president. He was previously a United States Senator from Illinois and a member of the Illinois State Senate." The entities and their labels are: "Barack Hussein Obama II" (Person, p), "August 4, 1961" (Date, d), "American" (Other, z), "the United States" (Loc, l), "January 20, 2009" (Date, d), "January 20, 2017" (Date, d), "Democratic Party" (Org, o), "African American" (Other, z), "United States Senator" (Other, z), "Illinois" (Loc, l), and "Illinois State Senate" (Org, o).

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3. Applications

- **Text categorization:** the assignment of predefined labels to target texts
 - Using a predefined set of parts of text categories



Ch. 9 – Semantic Disambiguation

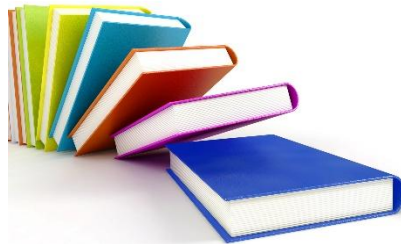
4. Traditional WSD

- WSD usually targets flat text
 - Underlines the process of computationally identifying the senses (i.e., semantic concepts designating the meanings) of words in context
 - To discover the author's intended meaning
- Aims at identifying the semantic meaning of data formulated with (possibly) different wordings and syntactic styles
 - Different from traditional text mining techniques (e.g., lexical pattern discovery, syntactic dependency, co-occurrence) which mainly capture the lexico-syntactic nature of text
 - They barely go beyond the surface appearance of words

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4. Traditional WSD

- The general WSD task consists of the following main elements:
 1. **Selecting words** for disambiguation
 2. Identifying and representing **word contexts**
 3. Using **reference knowledge** sources
 4. **Associating senses** with words
 5. Evaluating **semantic similarity** between senses



Ch. 9 – Semantic Disambiguation

4. Traditional WSD

4.1. Selecting Words for Disambiguation

- Two methods to select target words for disambiguation:
 - ***All-words***: the system is expected to disambiguate all words in a (flat) textual document
 - Although considered as a complete and exhaustive approach, yet the *all-words* approach remains extremely time-consuming
 - ***Lexical-sample***: specific target words are selected for disambiguation (usually one word per sentence)
 - Often the most ambiguous words, chosen using supervised learning methods trained to recognize salient words in sentences
 - More useful and efficient in practice

Ch. 9 – Semantic Disambiguation

4. Traditional WSD

4.2. Identifying and Representing Context

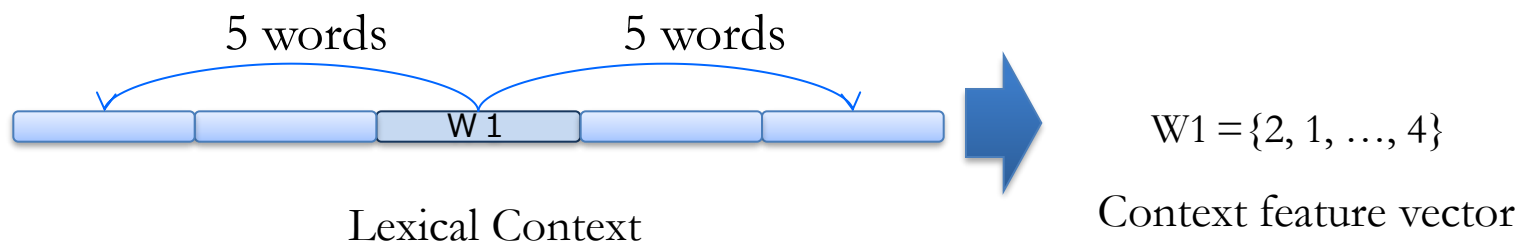
- WSD relies on the notion of **context** such that words that appear together in the same context usually have related meanings
- The **context of a word** in traditional flat textual data usually consists of the set of **terms in the word's vicinity**
 - Terms occurring to the left and right of the considered word, within a certain predefined window size
- Other features can also be used to describe context, such as information resulting from linguistic pre-processing:
 - Part-of-speech tags (e.g., verb, subject, etc.), grammatical relationships (e.g., verb-subject, verb-object, etc.)

Ch. 9 – Semantic Disambiguation

4. Traditional WSD

4.2. Identifying and Representing Context

- Once the context has been identified, it has to be effectively represented to perform disambiguation computations
- The traditional **bag-of-words paradigm** is broadly adopted with flat textual data
 - Where the context is processed as a set of terms surrounding the word to disambiguate



- A vector representation considering the number of occurrences of words in context can also be used

Ch. 9 – Semantic Disambiguation

4. Traditional WSD

4.3. Using Reference Knowledge Sources

- External knowledge is essential to perform WSD
 - We distinguish between **corpus-based** and **knowledge-based**
 - **Corpus-based:**
 - **Data-driven**, involving information about words previously disambiguated
 - Requires **supervised learning** from sense-tagged corpora (e.g., SemCor [124] and OntoNotes [143])
 - Where words/expressions have been associated with explicit semantic meaning, to enable predictions for new words

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4. Traditional WSD

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 - Requires **supervised learning** from sense-tagged corpora (e.g., SemCor¹ and OntoNotes²)
 - Where words/expressions have been associated with explicit semantic meaning, to enable predictions for new words
 - A *sense* here underlines a *labeled semantic category* to be used in supervised learning
 - It could be a formal *semantic concept* defined in a knowledge base (KB)

¹ https://www.gabormelli.com/RKB/SemCor_Corpus

² <https://catalog.ldc.upenn.edu/LDC2013T19>

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4. Traditional WSD

4.3. Using Reference Knowledge Sources

- External knowledge is essential to perform WSD
 - We distinguish between **corpus-based** and **knowledge-based**
 - **Corpus-based:**
 - A more restrictive view of *corpus-based* methods is as **word sense induction**
 - It seeks to automatically identify so-called *implicit senses*, i.e., *unlabeled semantic categories* (designating the *uses*) of target words in a raw (unlabeled) corpus (e.g., the Brown corpus ¹)
 - Methods in this category rely exclusively on corpus data where word relationships are derived from their contextualization and co-occurrence in the corpus itself
 - They cluster words with similar corpus statistics, producing an implicit sense inventory made of unlabeled word categories (clusters), where each category is an implicit sense

¹ https://www1.essex.ac.uk/linguistics/external/clmt/w3c/corpus_ling/content/corpora/list/private/brown/brown.html

Ch. 9 – Semantic Disambiguation

4. Traditional WSD

4.3. Using Reference Knowledge Sources

- External knowledge is essential to perform WSD
 - We distinguish between **corpus-based** and **knowledge-based**
 - **Corpus-based:**
 - While *corpus-based* methods have been popular in recent years, yet:
 - They are generally **data hungry**
 - They require **extensive training**
 - They require **huge text corpora**, or considerable amount of **manual effort** to produce a relevant sense-annotated corpus
 - Which are **not always available** or feasible in practice!
 - In addition, with *corpus-based* statistical approaches, the “true” understanding of words is hardly obtainable

¹ https://www1.essex.ac.uk/linguistics/external/clmt/w3c/corpus_ling/content/corpora/list/private/brown/brown.html

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4. Traditional WSD

4.3. Using Reference Knowledge Sources

- External knowledge is essential to perform WSD:
 - **Knowledge-based:**
 - Knowledge-driven, handling a structured (and explicit) sense inventory
 - Machine-readable KBs (dictionaries, thesauri, and/or lexical ontologies, e.g., WordNet¹, and Yago²) provide ready-made sources for knowledge-based WSD
 - Since words are evaluated according to their statistical distribution in a corpus, often capturing syntactic or stylistic factors instead of semantic meaning
 - Therefore, *knowledge-based* methods have been receiving more attention lately, and include most semi-structured disambiguation solutions

¹ <https://wordnet.princeton.edu/>

² <https://yago-knowledge.org/>

Ch. 9 – Semantic Disambiguation

4. Traditional WSD

4.4. Associating Senses with Words

- The culminating step in WSD is to associate senses with words
 - This is usually viewed as a word-sense classification task
 - We distinguish between ***supervised*** or ***unsupervised***
 - **Supervised** methods:
 - Involve the use of **machine-learning** techniques
 - A human expert manually annotates examples of words with the intended sense in context, where each sense underlines a *labeled semantic category*
 - » Provided as training data for a learning algorithm
 - The algorithm induces rules for assigning meanings to the words
 - » External knowledge (mainly *corpus-based*) is used
 - » Combined with the human experts' own knowledge of word senses when manually tagging the training examples

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4. Traditional WSD

4.4. Associating Senses with Words

- The culminating step in WSD is to associate senses with words
 - This is usually viewed as a word-sense classification task
 - We distinguish between *supervised* or *unsupervised*
 - **Unsupervised** methods:
 - Usually fully automated and do not require human intervention or training
 - Make use of a machine-readable knowledge bases (KBs, e.g., WordNet)
 - Identify the semantic concept (word sense), in the reference KB, that best matches the semantic concepts (word senses) of terms appearing in the context of the target word
 - Semantic concept matching is usually performed using a measure of semantic similarity between concepts in the reference semantic network

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4. Traditional WSD

4.4. Associating Senses with Words

- While supervised methods are effective, yet they require
 - A learning phase which is highly **time-consuming**
 - **A reliable training dataset** with a wide coverage
 - Which is **not always available!**
 - **Refer to corpus-based approaches in Section 4.3**
- In the following, we will mainly focus on:
 - **Unsupervised** and **knowledge-based** approaches

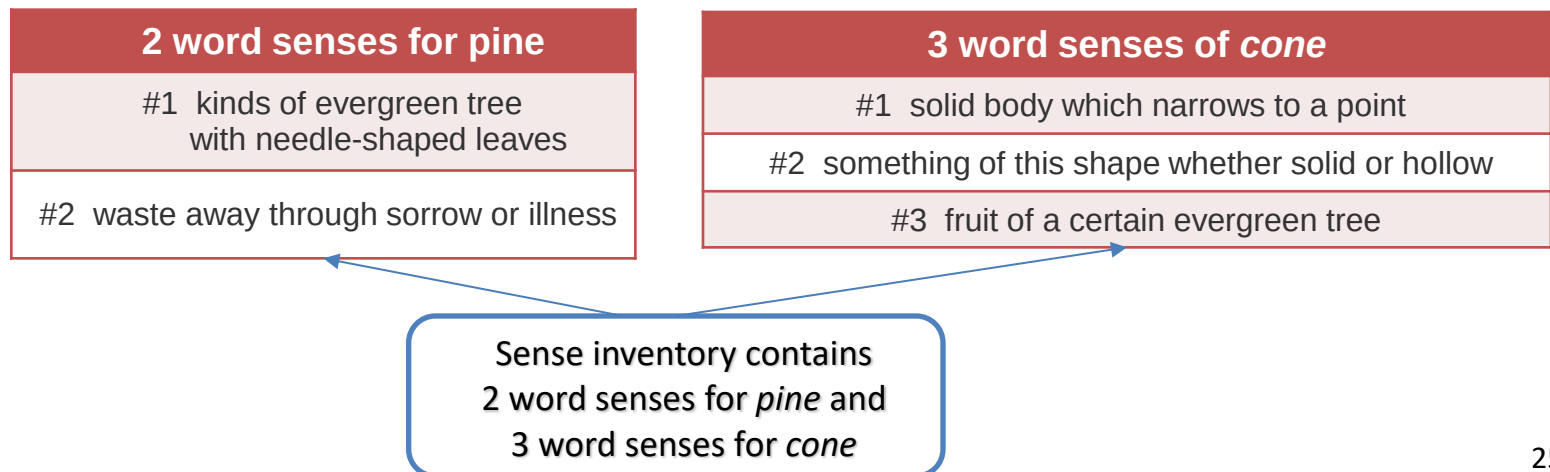


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4. Traditional WSD

4.4. Associating Senses with Words

- **Lesk's algorithm**
 - Created by Michael Lesk in 1986
 - Choose the word sense whose definition has maximum overlap with the definitions of each of the context words
- Example: Need to disambiguate keyword query **pine cone**

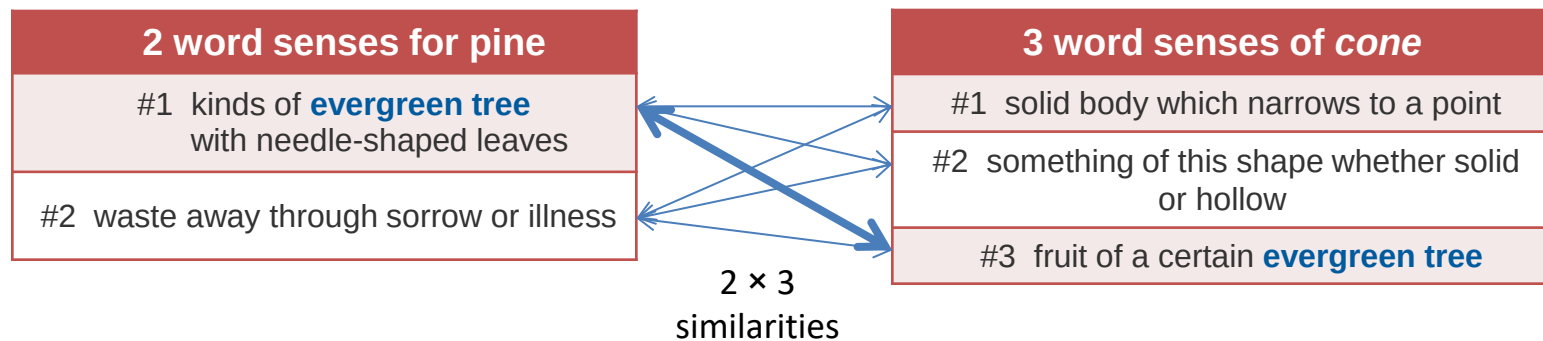


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4. Traditional WSD

4.4. Associating Senses with Words

- Example: Need to disambiguate keyword query **pine cone**



Using **gloss-based overlap** similarity:

- $\text{overlap}(\text{pine}\#1, \text{cone}\#1) = \text{overlap}(\text{pine}\#2, \text{cone}\#1) = \text{overlap}(\text{pine}\#1, \text{cone}\#2) = 0$
- $\text{overlap}(\text{pine}\#2, \text{cone}\#2) = \text{overlap}(\text{pine}\#2, \text{cone}\#3) = 0$
- $\text{overlap}(\text{pine}\#1, \text{cone}\#3) = \{\text{evergreen} + \text{tree}\} = 2$

⇒ **pine#1** and **cone#3** are selected!

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4. Traditional WSD

4.4. Associating Senses with Words

- Problem with Lesk's algorithm:
 - Combinatorial explosion of possible word sense combinations when more than two/three-word sentences are considered
- Example: "I saw a man who is 98 years old and can walk and tell jokes."
 - Several possible senses for ambiguous words: see(26), man(11), year(4), old(8), can(5), still(4), walk(10), tell(8), joke(3)

⇒ 43,929,600 sense combinations !!



Led to **Simplified Lesk** algorithm

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4. Traditional WSD

4.4. Associating Senses with Words

- **Simplified Lesk** algorithm
 - Disambiguating **target ambiguous words** in the input text
 - Regardless of the meaning of the other words occurring in the same context
 - **No need to perform disambiguation of all text!**
 - Correct meaning of each word is determined by finding the sense that leads to the highest **overlap between:**
 - Its **dictionary definition** and
 - The current **context of the word** in text

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4. Traditional WSD

4.4. Associating Senses with Words

○ Simplified Lesk algorithm

Example: The **bank** profits from the difference between the level of interest it pays for **deposits**, and the level of interest it charges in its **lending activities**.

- Target word: bank

2 × 1
similarities

2 word senses for bank	
#1	sloping land (especially the slope beside a body of water)
#2	a financial institution that accepts deposits and channels the money into lending activities

Using **gloss-overlap** similarity:

- $\text{overlap}(\text{bank\#1}, \text{context}) = 0$
- $\text{overlap}(\text{bank\#2}, \text{context}) = \{\text{deposits} + \text{lending} + \text{activities}\} = 3$

⇒ **bank#2** is selected!

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4. Traditional WSD

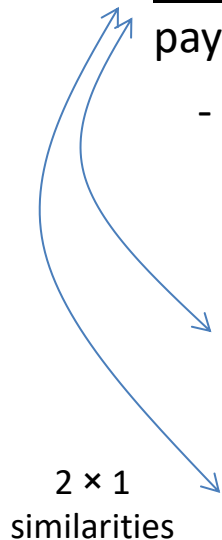
4.4. Associating Senses with Words

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⇒ **bank#2** is selected!

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5. WSD of Semi-structured Data

- Main challenges of semi-structured data processing:
 - Document **element/attribute** tag names and values are **subjectively defined**
 - **Heterogeneous** data **sources** could represent similar data
 - The same real world entity can be described using **different structures** and/or **tagging**

↳ Need to consider **structural & lexical properties**

⇒ **Semi-structured Semantic Ambiguity**

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Example 1: Ambiguity of tags

```
<star>  
  <first_name> James </first_name>  
  <last_name> Stewart </last_name>  
<star>
```

≠ senses

Celestial body?
Champion?
Plane figure with 5 points?
Lead actor?
Star network topology?



- ⇒ Different possible meanings/senses exist
- Following a lexical reference such as **WordNet**



Need for **context** information!

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Example 2: How to process context?

```
<?xml version="1.0"?>
<films>
  <picture id="301">
    <title> Rear Window </title>
    <director>
      <first_name> Alfred </firstName>
      <last_name> Hitchcock </last_name>
    </director>
    <year> 1964 </year>
    <cast>
      <star>
        <first_name> James </firstName>
        <last_name> Stewart </last_name>
      </star>
      <star>
        <first_name> Grace </firstName>
        <last_name> Kelly </last_name>
      </star>
    </cast>
  </picture>
</films>
```

Lead actor



Ch. 9 – Semantic Disambiguation

Example 3: Different tags and structures describing the same information...

```
<?xml version="1.0"?>
<films>
  <picture id="301">
    <title> Rear Window </title>
    <director>
      <first_name> Alfred </firstName>
      <last_name> Hitchcock </last_name>
    </director>
    <year> 1964 </year>
    <genre> Thriller </genre>
    <cast>
      <star>
        <first_name> James </firstName>
        <last_name> Stewart </last_name>
      </star>
      <star>
        <first_name> Grace </firstName>
        <last_name> Kelly </last_name>
      </star>
    </cast>
  </picture>
</films>
```

≠ ?

```
<?xml version="1.0"?>
<Movies>
  <Movie year= "1954">
    <Name> Rear Window </Name>
    <Directed_By>Hitchcock</Directed_By>
    <Actors>
      <Actor>
        <FirstName>Grace</FirstName>
        <LastName>Kelly</LastName>
      </Actor>
      <Actor>
        <FirstName>James</FirstName>
        <LastName>Stewart</LastName>
      </Actor>
    </Actors>
  </Movie>
</Movies>
```



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5. WSD of Semi-structured Data

- Core problem



How to resolve semi-structured semantic ambiguity



- Semi-structured element/attribute tag names and values

- **Synonymy:** several terms have the same meaning
- **Hyponymy:** a term has several meanings
- **Metonymy:** a term implies others
- **Hypernymy:** a term inherits the meaning of another
- **Derivation:** a term is derived from another
- Etc.



How to process semi-structured context



- Considering document structure

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5. WSD of Semi-structured Data

- Goal: Performing semi-structured semantic analysis and disambiguation
- Main objectives:
 - **Augmenting semi-structured** syntactic information
 - With a **semantic description** from a lexical reference
 - I.e., Wordnet commonly used for WSD
 - Taking into account the **structural properties** of the data
 - Allowing the user to **fine-tune** the process
 - Following her needs



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5. WSD of Semi-structured Data

5.1. Limitations of Traditional Solutions

- Few approaches target semi-structured semantic-aware analysis
 - Extended from traditional flat text WSD
- Motivation 1: Ignoring *semantic ambiguity*
 - Running disambiguation on all semi-structured nodes (*all-words*)
 - ⇒ Time consuming and needless
 - Rather than only processing those nodes which are most ambiguous (*lexical-sample*)



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- Motivation 2: Partially considering *structure*

```
<?xml version="1.0"?>
<films>
  <picture id="301">
    <title> Rear Windom </title>
    <director>
      <first_name> Alfred </firstName>
      <last_name> Hitchcock </last_name>
    </director>
    <year> 1964 </year>
    <genre> Thriller </genre>
    <cast>
      <star>
        <first_name> James </firstName>
        <last_name> Stewart </last_name>
      </star>
    </cast>
  </picture>
</films>
```



Context?



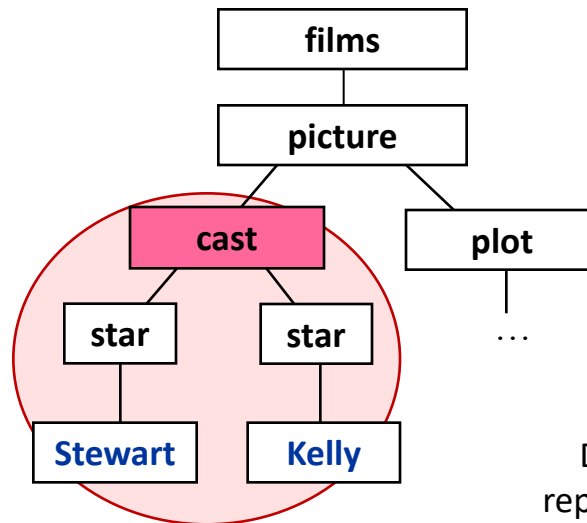
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- Motivation 2: Partially considering *structure*
 - Different context models have been considered:
 - **Parent node** context [1, 2]
 - **Root-path** context [3, 4]
 - **Sub-tree** context [5]
 - **Versatile** structure context [6]
- } **Limited coverage**
- Identifying ***crossable edges*** using a *Gaussian decay function*
 - ⇒ Convoluted and **intricate mathematical model**
 - Designed for **general structured data**



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- Motivation 3: Neglecting structure in *processing context*
 - Using the **bag-of-words** paradigm
 - Which considers a set of equivalent context nodes
 - Yet, structure affects the impact of context nodes
 - Based on their position in the document tree



DOM tree
representation

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5. WSD of Semi-structured Data

5.1. Limitations of Traditional Solutions

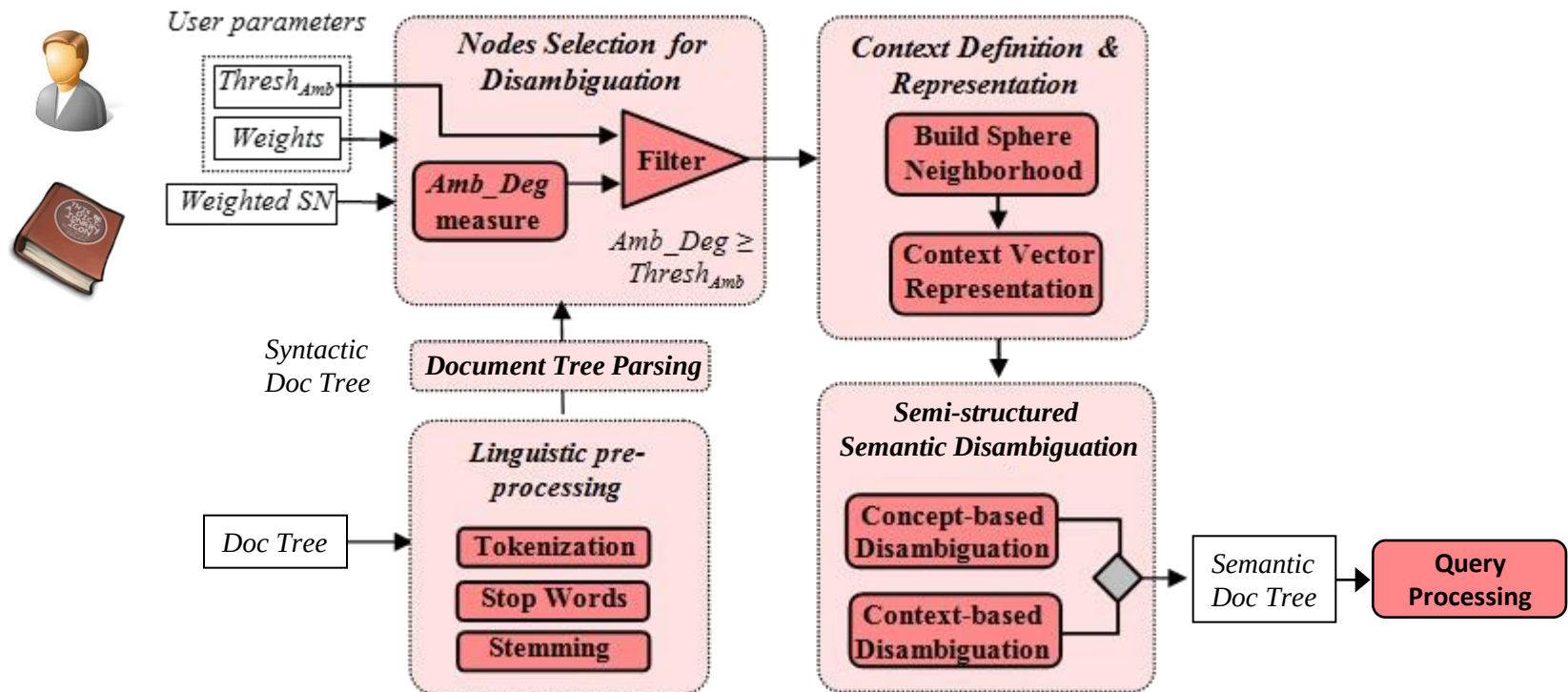
- Motivation 4: Existing methods are mostly *static*
 - Adopting a fixed context size (e.g., parent node, or root path)
 - Using preselected semantic similarity measures (e.g., edge-based measure, or gloss-based measure)
 - ⇒ **User involvement**/system adaptability **is minimal**



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5. WSD of Semi-structured Data

5.2. Proposed Solution



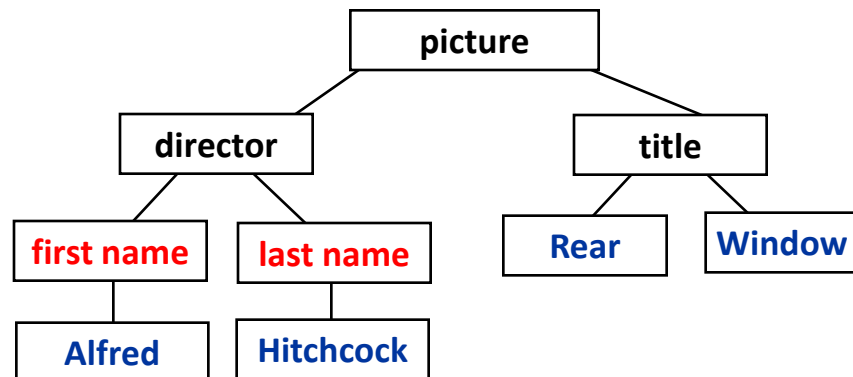
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5. WSD of Semi-structured Data

5.2. Proposed Solution

5.2.1. Linguistic Pre-processing

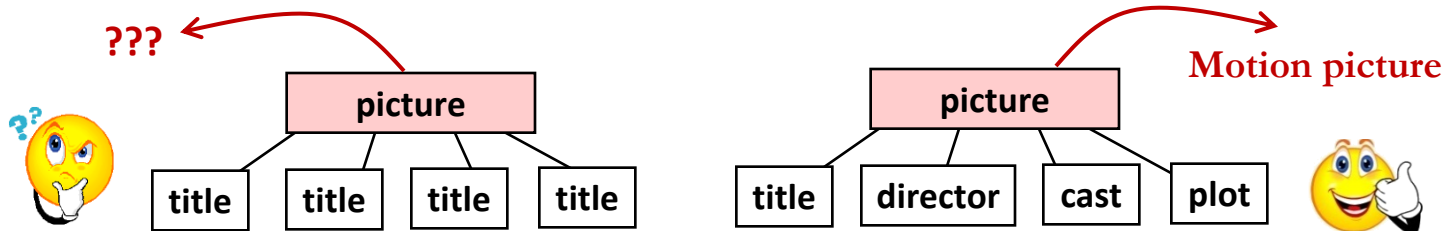
- General process: i) tokenization, ii) stop word removal, iii) stemming
- We consider 3 possible inputs:
 - Element/attribute **tag names** consisting of **individual words**
 - Element/attribute **tag names** consisting of **compound words**
 - Usually made of two individual terms (t_1 and t_2)
 - Element/attribute **text values**



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5.2.2. Node Selection of Disambiguation

- Quantifying the notion of semi-structured node **ambiguity degree**:
 - Based on a set of assumptions:
 - **Polysemy**: Nodes having labels with multiple senses (in WordNet) are considered ambiguous
 - **Depth**: Nodes closer to the root usually refer to general concepts, and can thus be considered ambiguous
 - **Fan-out**: Nodes with few number of distinct children node labels are considered ambiguous



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5.2.2. Node Selection of Disambiguation

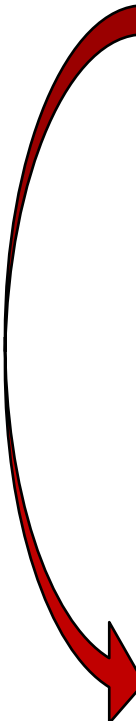
- Defining an **ambiguity degree measure** of a semi-structured node:

– Combination of three factors:

□ **Polysemy:** $Amb_{Polysemy}(x.l, SN) = \frac{senses(x.l) - 1}{Max(senses(SN)) - 1} \in [0,1]$

□ **Depth:** $Amb_{Depth}(x, T) = 1 - \frac{x.d}{Max(depth(T))} \in [0,1]$

□ **Fan-out:** $Amb_{Density}(x, T) = 1 - \frac{\overline{x.f}}{Max(fan-out(T))} \in [0,1]$



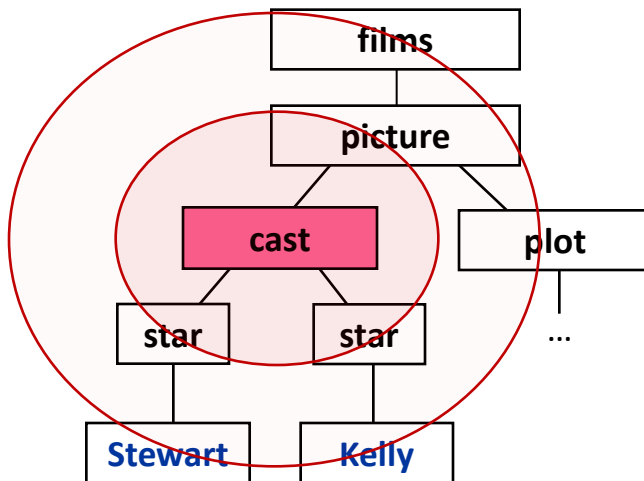
$$Amb_Deg(x, T, SN) = \frac{w_{Polysemy} \times Amb_{Polysemy}(x.l, SN)}{w_{Depth} \times (1 - Amb_{Depth}(x, T)) + w_{Density} \times (1 - Amb_{Density}(x, T)) + 1} \in [0,1]$$

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5.2.3. Context Definition and Representation

5.2.3.1. Introducing the **sphere neighborhood** context model

- Inspired from the *sphere-search* paradigm in semi-structured IR [7]
- Considering the whole structural surrounding of a node
 - Including **ancestors**, **descendants**, and **siblings**



Semi-structured ring at distance d

$$R_d(x) = \{x_i \in T \mid \text{Dist}(x, x_i) = d\}$$



Semi-structured sphere at distance d

$$S_d(x) = \{x_i \in T \mid x_i \in R_{d_j}(x) \wedge d_j \leq d\}$$

Ch. 9 – Semantic Disambiguation

5.2.3. Context Definition and Representation

5.2.3.2. Producing context vector representation

- Evaluate the impact of each of the context nodes
 - In performing semantic disambiguation
 - In contrast with ***bag-of-words*** paradigm
- Quantified based on 2 assumptions:
 - **Closeness:** semi-structured context nodes closer to the target node should better influence the latter's disambiguation
 - **Frequency:** Nodes with identical labels, occurring multiple times in the context of a target node, should better influence the latter's disambiguation

Ch. 9 – Semantic Disambiguation

5.2.3. Context Definition and Representation

5.2.3.2. Producing context vector representation

- Defining a context vector weights

– Combination of two factors:

□ **Closeness:** $\text{Struct}(x_i, S_d(x)) = 1 - \frac{\text{Dist}(x, x_i)}{d+1}$

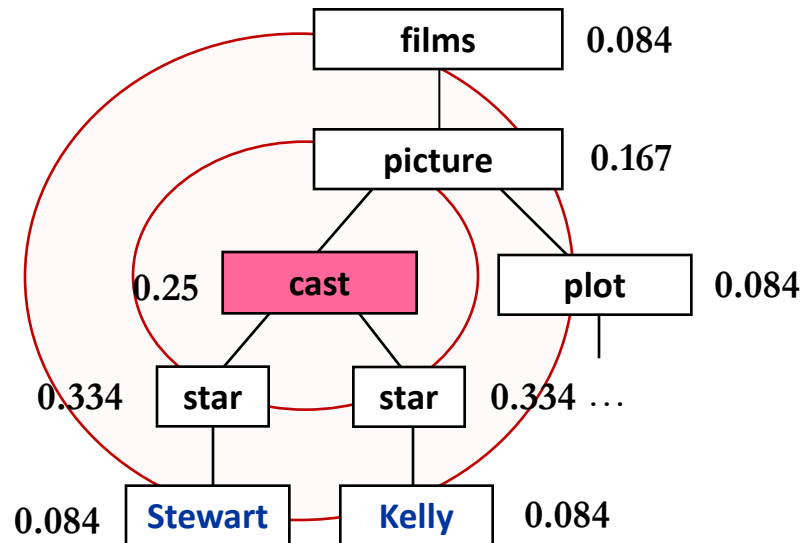
□ **Frequency:** $\text{Freq}(l_r, S_d(x)) = \sum_{x_i \in S_d(x) / x_i.l = l_r} \text{Struct}(x_i, S_d(x))$

$w_{\overrightarrow{V_d(x)}}(l_r) = \frac{\text{Freq}(l_r, S_d(x))}{\text{Max}_{\text{Freq}}} \in [0,1]$

Ch. 9 – Semantic Disambiguation

5.2.3. Context Definition and Representation

5.2.3.2. Producing context vector representation


 $\overrightarrow{V_2(T[2])}$

<i>Cast</i>	<i>Picture</i>	<i>Star</i>	<i>Films</i>	<i>Stewart</i>	<i>Kelly</i>	<i>Plot</i>
0.25	0.1667	0.3334	0.0835	0.0835	0.0835	0.0835

Ch. 9 – Semantic Disambiguation

5.2.4. Semi-structured Semantic Disambiguation

- Combining two approaches: *concept-based* and *context-based*

- Concept-based

- Comparing the target node senses with context nodes senses
 - Using a combination of semantic similarity measures
 - » Applied on the reference semantic network (WordNet)
 - » *Edge-based, Node-based, and Goss-based*



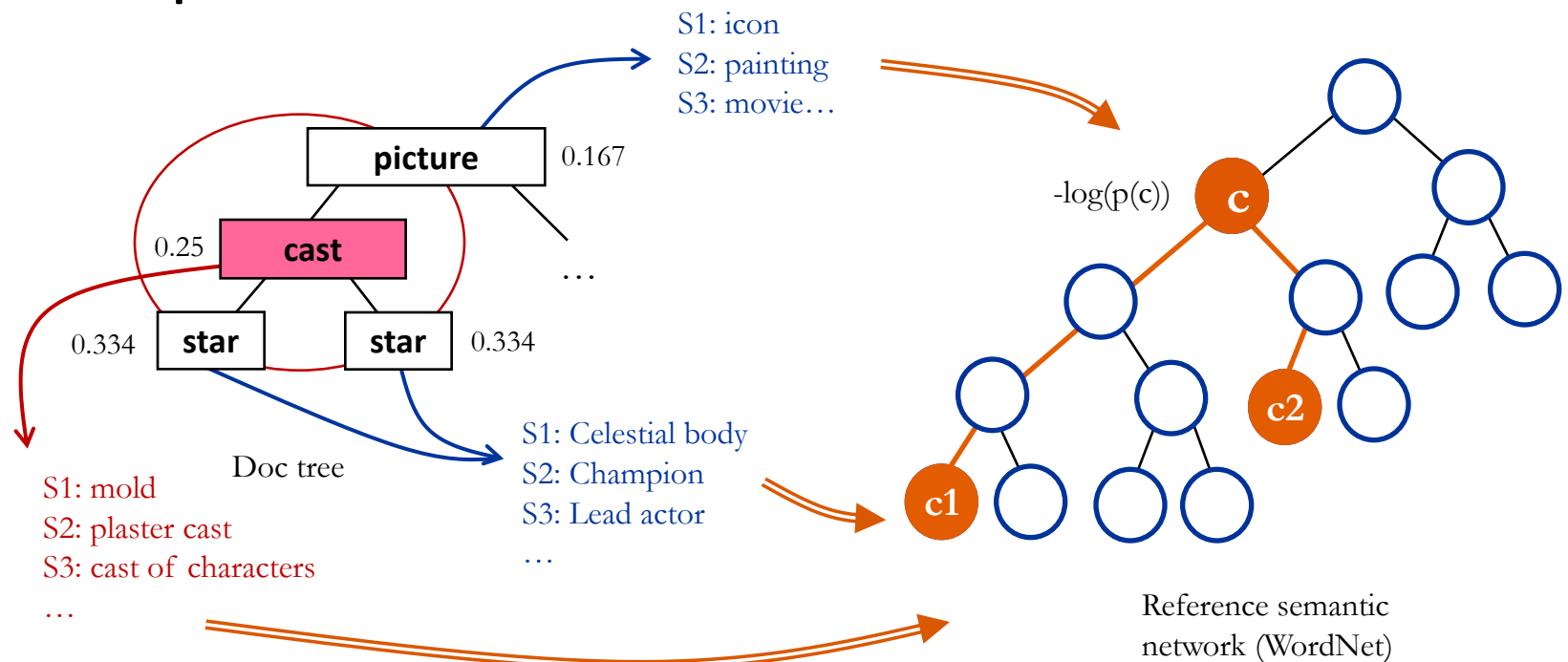
$$\text{Concept_Score}(s_p, S_d(x), SN) = \sum_{x_i \in S_d(x)} \frac{\text{Max}_{s_j^i \rightarrow x_i.l} \left(\text{Sim}(s_p, s_j^i, \overline{SN}) \right) \times w_{\overline{V_d(x)}}(x_i.l)}{|S_d(x)|} \in [0,1]$$

Ch. 9 – Semantic Disambiguation

5.2.4. Semi-structured Semantic Disambiguation

- Combining two approaches: concept-based and context-based

- Concept-based



Ch. 9 – Semantic Disambiguation

5.2.4. Semi-structured Semantic Disambiguation

- Combining two approaches: concept-based and context-based

- **Context-based**

- Comparing the target node sphere neighborhood in the doc tree
 - With each of its sense (concept) sphere neighborhoods in the reference semantic network (WordNet)



» Using vector comparison measures, e.g., *cosine*

$$\text{Context_Score}(s_p, S_d(x), \text{SN}) = \cos(\overrightarrow{V_d(x)}, \overrightarrow{V_d(s_p)}) \in [0, 1]$$

$\overrightarrow{V_d(x)}$: Sphere context vector of node in doc tree

$\overrightarrow{V_d(s_p)}$: Sphere context vector of sense s_p of node in SN

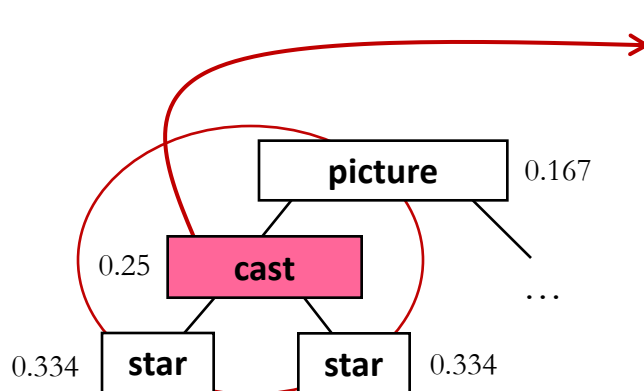
Ch. 9 – Semantic Disambiguation

5.2.4. Semi-structured Semantic Disambiguation

- Context-based

$$\overrightarrow{V_d(s_2)}$$

<i>Cast</i>	<i>Bandage</i>	<i>Patch</i>	...
0.25	0.1232	0.2233	...

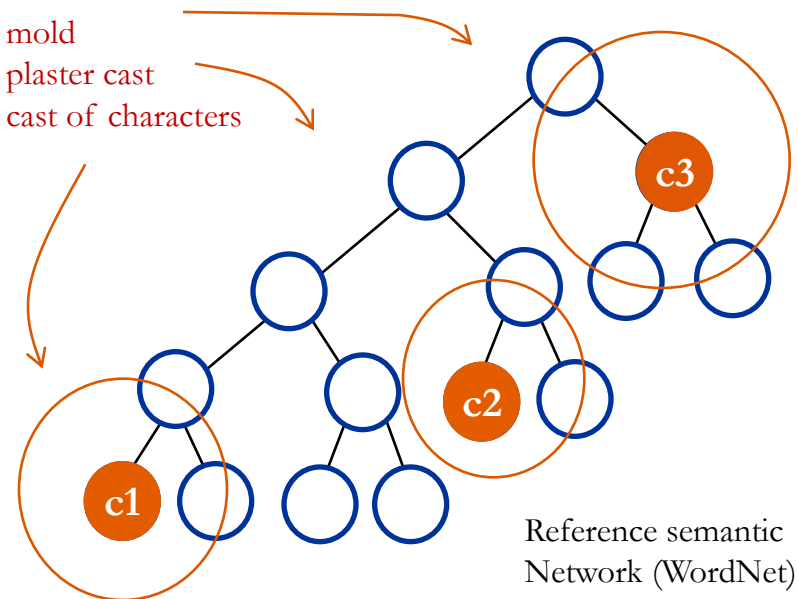


Doc tree

$$\overrightarrow{V_d(x)}$$

<i>Cast</i>	<i>Picture</i>	<i>Star</i>	...
0.25	0.1667	0.3334	...

S1: mold
S2: plaster cast
S3: cast of characters
...

Reference semantic
Network (WordNet)
$$\overrightarrow{V_d(s_1)}$$

<i>Cast</i>	<i>Mold</i>	<i>Stamp</i>	...
0.25	0.1544	0.1544	...

$$\overrightarrow{V_d(s_3)}$$

<i>Cast</i>	<i>Gathering</i>	<i>Ensemble</i>	...
0.25	0.1566	0.1032	...

Ch. 9 – Semantic Disambiguation

5. WSD of Semi-structured Data

5.2. Proposed Solution

5.2.4. Semi-structured Semantic Disambiguation

- Complexity of *concept-based* approach:

$$O(|senses(x.\text{?})| \times |S_d(x)| \times |senses(x_i.\text{?})| \times |SN|)$$

- Complexity of *context-based* approach

$$O(|senses(x.\text{?})| \times \underbrace{(|S_d(x)| + |S_d(s_p)|)})$$

⇒ **More efficient**

Cosine similarity
executed in
average linear time

Ch. 9 – Semantic Disambiguation

5. WSD of Semi-structured Data

5.2. Proposed Solution

5.2.4. Semi-structured Semantic Disambiguation

- Combining concept-based and context-based approaches

$$\begin{aligned} &\text{Concept_Score}(s_p, S_d(x),) = \\ &w_{\text{Concept}} \times \text{Concept_Score}(s_p, S_d(x), \text{SN}) \\ &\quad + \\ &w_{\text{Context}} \times \text{Context_Score}(s_p, S_d(x), \text{SN}) \end{aligned}$$



User can fine-tune disambiguation process

- Considering the nature of semi-structured data being processed

Ch. 9 – Semantic Disambiguation

6. Challenges and Ongoing Directions

6.1. *Using collaborative knowledge sources*

- **Conventional KBs** also **show** many **limitations**:
 - Need for significant human effort to create and maintain
 - Need for wide coverage which is difficult to achieve manually from experts
 - Need to update information in the KB in a timely manner which cannot always be achieved by experts
- Increasing need to consider **non-conventional sources** such as Wikipedia, Google, Yahoo, and Flickr, which are **collaboratively built**
 - Where content can be efficiently created and updated by non-experts from different domains, and then the produced content is simply vetted by experts
 - This requires significantly less effort than manually creating the information from scratch, and allows for a wider coverage and faster update

Ch. 9 – Semantic Disambiguation

6. Challenges and Ongoing Directions

6.1. *Using collaborative knowledge sources*

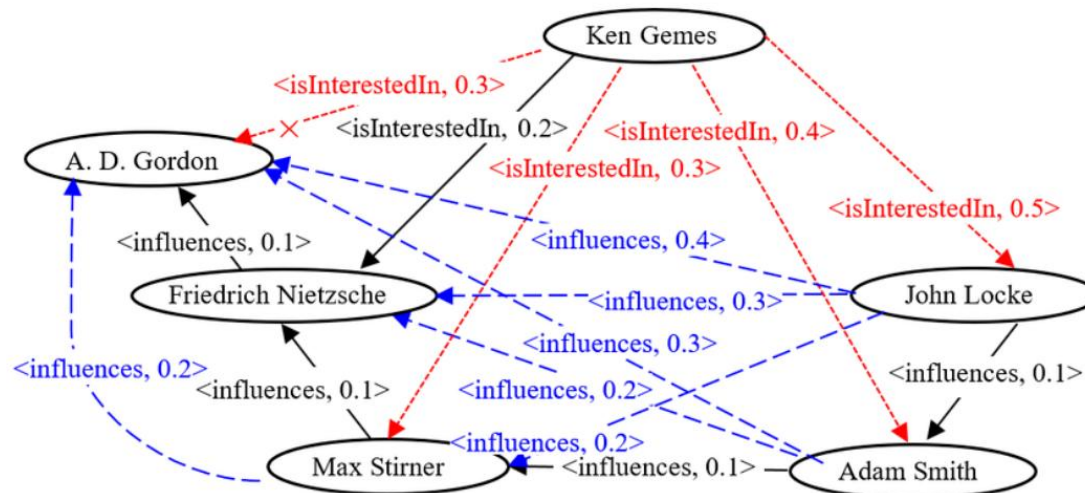
- While, such sources are semi-structured in nature, nonetheless information is **not fully formalized** or disambiguated
 - For instance, Wikipedia's text corpus is partially structured in pages (articles) and info-boxes , with various relationships among pages, e.g.:
 - redirection pages, internal hyperlinks, inter-language links, as well as category pages
 - Yet, Wikipedia pages also contain **vast amounts of unstructured text** with no linkage to any pages or other terms (i.e., with no semantic mean-ing)
 - Thus, dedicated additional processing is required, e.g.:
 - mapping ambiguous/related terms/pages together and identifying missing relationships between terms/pages, in order to perform WSD

Ch. 9 – Semantic Disambiguation

6. Challenges and Ongoing Directions

6.1. Using collaborative knowledge sources

- Expanding/integrating semi-structured sources such as Wikipedia, with structured KBs such as WordNet, to fill-in the semantic knowledge gap, has been a promising research direction
 - with successful projects such as **Yago** and **DBpedia**



A fragment of fuzzy knowledge graph extended from YAGO

Ch. 9 – Semantic Disambiguation

6. Challenges and Ongoing Directions

6.2. *Using social knowledge sources*

- Another form of collaboratively built knowledge sources has emerged in the Social Semantic Web, known as **folksonomies**
- A folksonomy is basically a 3-dimensional data structure (usually represented as a 3-uniform graph) whose dimensions are represented by **users**, **tags**, and **resources**
 - users assign tags to Web resources, without a reference KB
 - Then a post-processing, organization, and mining of the tags and their relationships with users and resources would identify user vocabularies describing the resources
 - Such vocabularies are commonly referred to as **emergent semantics**



a (bottom-up) complement to the more formalized (top-down) knowledge organization in conventional KBs

Ch. 9 – Semantic Disambiguation

6. Challenges and Ongoing Directions

6.2. *Using social knowledge sources*

- Producing useful folksonomies requires dedicated processing to map the semi-structured and ambiguous tags in the folksonomy with concepts in the KB
 - Extending the KB with new concepts/relationships highlighting the users' semantics
 - Then, either the user vocabulary or the extended KB can be used (with corpus-based and/or knowledge-based analysis) to achieve quality WSD
- NOTE: processes originally developed for semi-structured disambiguation can be utilized to handle collaboratively built semi-structured sources
 - and vice versa: techniques devised to disambiguate concepts and identify relationships in semi-structured knowledge sources like Wikipedia or folksonomies could inspire the development of new XML disambiguation techniques

Ch. 9 – Semantic Disambiguation

6. Challenges and Ongoing Directions

6.3. Using Implicit Semantics

- Another direction to improve XML semantic analysis and disambiguation is to incorporate **implicit semantics** (a.k.a. **latent semantics**) inferred from the statistical analysis of XML element tag names/text values
 - following the basic idea that: documents which have many node labels in common are semantically closer than ones with fewer node labels in common
- *Implicit concepts* are *synthetic* concepts generated by **extracting latent relationships** between terms in a document (or a document collection)
 - or by **calculating probabilities** of encountering terms
 - generated concepts do not necessarily align with any human-interpretable concept
 - This is different from conventional concept-based semantic analysis, which utilizes *explicit concepts* representing *real-life* entities/notions defined following human perception
 - concepts defined within a KB such as WordNet or Wikipedia

Ch. 9 – Semantic Disambiguation

6. Challenges and Ongoing Directions

6.3. Using Implicit Semantics

- Yet, existing implicit semantic analysis methods suffer from various limitations:
 - implicit concepts are difficult to understand and evaluate by human users
 - the number of generated implicit concepts depends on statistical/algebraic analysis rather than the actual meaning of the data
 - existing methods are relatively complex and do not scale well
 - to be further explored toward handling semi-structured data...



Deep Language Models are providing promising solutions...



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